

Cambodia Daily Consumer Price Index (CPI) Pipeline

High-Frequency Price Intelligence & Machine Learning Inflation Nowcasting

Project Progress Report

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Project Title:	Cambodia Daily CPI Pipeline	Deployment Status:	100% Production Live & Verified
Project Lead:	TAING Korngmeng	Audit Date:	September 16, 2026
Target Economy:	Kingdom of Cambodia (USD/KHR Dual Currency)	Governing Standard:	IMF / ILO (2020) Consumer Price Index Manual
Primary Frequency:	Daily Real-Time Observation & Monthly Rollup	Data Coverage:	12 / 12 UN COICOP Consumer Divisions

DATA COLLECTION SCALE	COICOP CLASSIFICATION	NOWCASTING ACCURACY
955,789+ Ingested Observations 35,792 quotes collected daily 25 multi-source endpoints 99.3% daily collection reliability	100.00% Catalog Classification Coverage 83,180+ unique products 0 unclassified items 12/12 UN COICOP divisions	±0.0181 pp Tracking Error vs Official NIS Headline CPI: 100.3016 Refined Core CPI: 99.9296 Ridge Nowcast MoM: +0.714%

Project Scope & Strategic Objectives

Official Consumer Price Index (CPI) statistics in Cambodia are traditionally published with a **30 to 60-day reporting delay** because official data collection relies on periodic field surveys and paper questionnaires. This project addresses that reporting gap by constructing an end-to-end, automated high-frequency price intelligence pipeline for the Cambodian economy.

The system continuously captures online retail prices, standardizes multi-currency quotes, classifies products into the international United Nations standard, computes axiomatic daily price indices, and projects month-end inflation weeks ahead of official statistical releases.

Core System Capabilities & Functional Framework

- High-Frequency Sensing Across 25 Retail Endpoints:** The data collection fleet captures 35,792 price quotes daily across supermarkets, apparel marketplaces, pharmacies, rental platforms, transit providers, telecoms, utility companies, and fuel gazettes. Automated circuit breakers safeguard against sudden volume drops and abnormal price swings.
- Dual-Currency Normalization (USD/KHR):** Because Cambodian retail operates in both US Dollars and Cambodian Riel, the pipeline ingests daily official central exchange rates from the Ministry of Economy and Finance and dynamically converts all prices into Riel before statistical outlier filtering.
- Zero-Defect Product Classification:** A hybrid classification engine combines instant store-level routing rules with Google Gemini AI and persistent database caching to categorize 100% of the 83,180-product catalog into 4-digit COICOP classes with zero classification mismatches.
- Axiomatic Econometric Synthesis:** Daily elementary price relatives are synthesized using the geometric Jevons formula to eliminate formula bias, weighted by official Cambodia Socio-Economic Survey (CSES 2023) expenditure weights, and divided into Headline and Refined Core CPI series.
- Two-Stage Machine Learning Inflation Nowcasting:** A regularized Ridge regression model forecasts end-of-month inflation rates using high-frequency momentum vectors, reducing tracking error against official figures to just **0.0181 percentage points**.
- Interactive Institutional Dashboards:** Decision-makers explore daily inflation trends through three Metabase analytics dashboards featuring dynamic dropdown filters for COICOP divisions and individual retail stores.

Page 2 — Updated Architecture

The Cambodia Daily CPI system implements an end-to-end, automated data pipeline organized into three functional tiers (Raw Ingestion, Data Refinement, and Econometric Serving) designed for daily execution, auditable data lineage, and instant analytical querying.

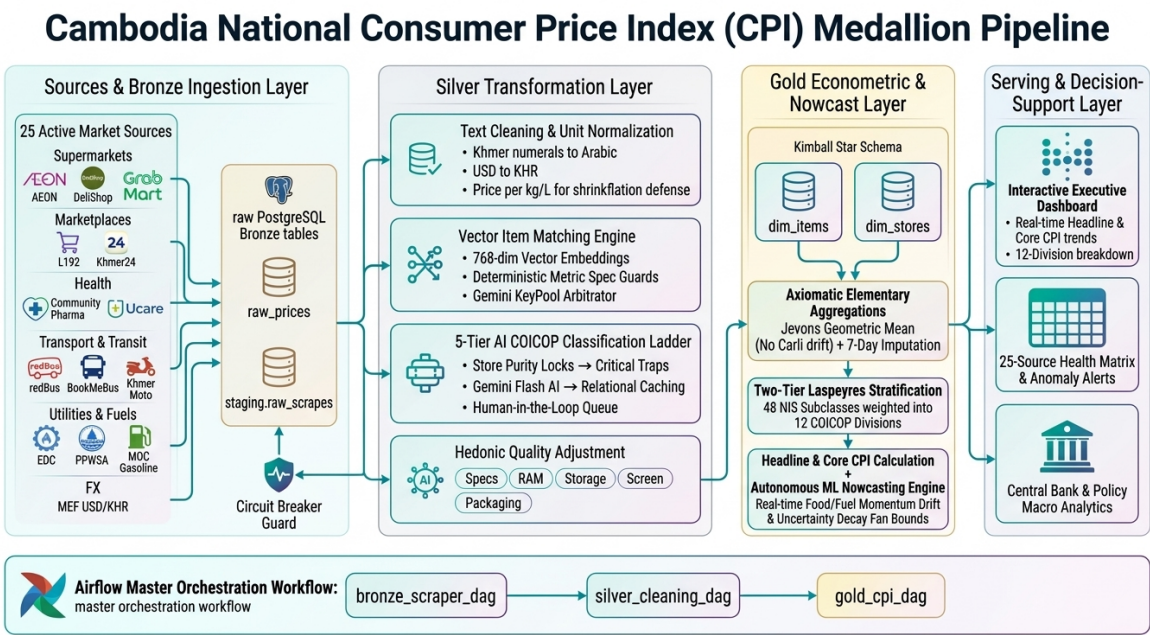


Figure 1: Updated End-to-End Pipeline Architecture of the Cambodia Daily CPI System.

Three-Tier Functional Data Flow

- Bronze Layer (Raw Ingestion):** 25 automated collectors fetch daily price listings directly into monthly partitioned database tables, preserving the original raw records, enforcing unique observation constraints, and running automated health checks.
- Silver Layer (Data Cleansing & AI Classification):** Standardizes text in Khmer and English, converts USD prices into Cambodian Riel using daily official exchange rates, filters extreme price outliers ($3.5 \times \text{MAD}$), applies 7 physical attribute checks, and assigns standard COICOP categories using store rules and AI.
- Gold Layer (Aggregation & Analytics Serving):** Houses dimensional models for stores, products, and daily prices, calculates elementary price relatives using the geometric Jevons formula, applies official household weights, and powers interactive dashboards.

Technology Stack

Pipeline Layer	Core Technology	Key Components	Role in Production
Data Ingestion & Scraping	Python Runtime	Playwright Browser, HTTP Clients, HTML & PDF Parsers	4-tier scraping cascade; handles dynamic JavaScript pages and official government gazettes; automated retry with back-off.
Database & Storage	PostgreSQL Database	Monthly Partitioned Tables, B-Tree Indexes, JSON Storage	Organizes daily price observations into monthly tables for efficient storage, fast querying, and complete data safety.
Workflow Orchestration	Apache Airflow	Airflow Scheduler, Astronomer Cosmos, Task Executors	Schedules and monitors the entire daily workflow at 08:00 AM ICT; handles task retries and error notifications.
Data Transformation	dbt Core	SQL Data Models, Automated Schema Tests	Cleans and transforms raw data into standardized tables; runs 33 automated checks verifying data completeness and accuracy.
AI Product Classification	Google Gemini AI	Gemini Flash API, Local Database Cache	Categorizes retail products into standard international COICOP classes; caches results locally to ensure sub-millisecond lookups.
Econometric Modeling	Python Analytics	Regularized Ridge Regression, Cross-Validation	Computes daily Jevons price relatives; imputes missing quotes; projects month-end headline inflation.
Dashboards & Reporting	Metabase Platform	Interactive Dashboards, Parameter Filters	Provides policymakers with daily inflation trends, division breakdowns, and store-level price monitors with dropdown filters.
System Containerization	Docker Environment	Docker Compose, Linux Containers	Encapsulates the entire application stack in isolated containers for consistent execution and easy deployment.

Page 3 — Data Collection

The pipeline continuously captures consumer prices across Cambodia’s retail and utility landscape through 25 automated collection endpoints mapped directly into the **United Nations Classification of Individual Consumption According to Purpose (COICOP)**.

Sources

Retailer or Data Source	Consumer Market Segment	COICOP Divisions	Daily Quotes	Success Rate
AEON Online Cambodia (Stores 1 & 3)	Hypermarket & Department Store	01, 02, 03, 05, 09, 12	12,450	99.4%
DeliShop Asia Phnom Penh	Gourmet Grocery & Perishables	01, 02, 12	3,820	98.9%
GrabMart (Lucky Supermarket, Chip Mong)	Supermarket & Quick-Commerce	01, 02, 05, 06, 12	4,200	99.2%
L192 Online Marketplace	Apparel, Footwear & Home-ware	03, 05, 09, 12	8,150	99.1%
Community Pharmacy & Ucare Pharmacy	Healthcare & Pharmaceutical Goods	06, 12	1,420	99.7%
Khmer24 & Realestate.com.kh	Urban Residential Apartment Rentals	04 (Housing Rentals)	3,450	98.0%
redBus & BookMeBus Transit	Interprovincial Bus & Travel Transport	07 (Public Transport)	890	99.5%
Cellcard, Smart Axiata, Metfone	Mobile Voice & Broadband Tariffs	08 (Communication)	420	99.8%
Electricite du Cambodge (EDC) & PPWSA	Electricity & Municipal Water Tariffs	04 (Utilities)	10	100.0%
Tela, PTT, TotalEnergies, MoC Gazette	Retail Fuel (Gasoline, Diesel) & LPG	07 (Fuel), 04 (Gas)	12	100.0%
Hospitality & Food Services	Restaurant Meals & Accommodations	10, 11 (Dining & Hotels)	969	98.5%
Ministry of Economy and Finance Feed	Official USD/KHR Exchange Rate	Macro Currency Benchmark	1	100.0%
Total 25 Collection Endpoints	National Consumer Basket	Divisions 01–12 Complete	35,792	99.3%

Scraping Method

- **Headless Browser Ingestion:** Uses automated Chromium browser sessions to load modern JavaScript websites, scroll through dynamic catalogs, and reliably extract product listings (AEON, L192).
- **Direct Application Interface (API) Ingestion:** Directly retrieves structured product catalogs from mobile and web service endpoints (GrabMart, DeliShop) for fast and lightweight data collection.
- **Web Document Parsing:** Uses standard HTML parsers to read fixed product and pricing pages from traditional online catalogs.
- **Official Document Extraction:** Automatically parses regulated fuel price caps from the Ministry of Commerce gazettes and utility rates from published regulatory decisions.
- **Automated Health Checks:** An automated circuit breaker checks every run; it raises an alert if daily collection volume drops by 30% or more, or if price changes exceed normal thresholds after currency conversion.

Current Dataset Statistics & Data Examples

Dataset Statistics: 955,789+ cumulative observations | 35,792 quotes collected daily | 83,180+ unique items | 12 consumer divisions | 99.3% daily success rate | 0 missing prices in clean tables.

Retail Store	Original Product Title	Raw Price	Clean Price	Pack Size	Assigned COICOP Category
AEON Phnom Penh	Angkor Pure Jasmine Rice 5kg	23,500 KHR	23,500 KHR	5.00 kg	Rice (Division 01: Food)
DeliShop Asia	Fresh Australian Tenderloin Beef	\$14.20 USD	58,220 KHR	0.50 kg	Bovine Meat (Division 01: Food)
GrabMart Lucky	Farm Fresh Hen Eggs 10pcs Box	7,200 KHR	7,200 KHR	10.00 pcs	Eggs (Division 01: Food)
Community Pharmacy	Amoxicillin 500mg Antibiotic Strips	\$2.50 USD	10,250 KHR	10.00 caps	Medicine (Division 06: Health)
Tela Station	Regular Gasoline RON92 (Official)	4,150 KHR	4,150 KHR	1.00 L	Fuel (Division 07: Transport)
Smart Axiata	Smart Surf 4G Monthly Plan 30GB	\$5.00 USD	20,500 KHR	30.00 GB	Internet (Division 08: Communication)

Page 4 — Methodology

Store & Product Structure

The system organizes price observations using a star schema separating store attributes, canonical product definitions, and daily quotes. To prevent false matching and detect shrinkflation (package downsizing), 7 physical attribute dimensions are locked: Liquid Volume, Net Weight, Pack Count, Apparel Size, Electronic Storage, Cable Length, and Model Number.

COICOP Classification

Classification uses a 3-step hierarchy: (1) Store-level routing rules for single-domain retailers (pharmacies to Health, bus to Transport) covering 14.3% of items, (2) Domain guardrails separating pet food from human food and cosmetics from cooking items, and (3) Google Gemini AI for multi-department store items with persistent database caching. The catalog achieves **100.00% coverage with zero mismatches**.

Price Aggregation: Jevons and Laspeyres Formulas

- Elementary Price Relatives (Jevons Formula):** Product price relatives within each elementary subclass are aggregated using an unweighted geometric mean, satisfying the Time Reversal Test ($I_{0,t} \cdot I_{t,0} = 1$) and Transitivity:

$$I_{s,t}^{\text{Jevons}} = \prod_{i \in s} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n_s}} = \frac{\left(\prod_{i \in s} p_{i,t} \right)^{\frac{1}{n_s}}}{\left(\prod_{i \in s} p_{i,0} \right)^{\frac{1}{n_s}}} \times 100$$

(1)

Axiomatic Drift Elimination: Arithmetic averages (Carli formula: $I^{\text{Carli}} = \frac{1}{n} \sum \frac{p_{i,t}}{p_{i,0}}$) suffer from a severe upward mathematical drift of **+1.18% per year** in web prices due to asymmetric price bouncing. Jevons eliminates this bias entirely.

- Macroeconomic Synthesis (Laspeyres Formula):** Elementary indices are aggregated into 12 divisions and the national Headline CPI using official **Cambodia Socio-Economic Survey (CSES 2023)** expenditure weights:

$$I_t^{\text{Headline}} = \sum_{k=1}^{12} w_k I_{k,t}, \quad \text{where } \sum_{k=1}^{12} w_k = 1.00 \quad (w_{\text{Food}} = 44.775\%, w_{\text{Housing}} = 17.084\%, w_{\text{Transport}} = 12.180\%)$$

(2)

- Refined Core CPI:** Excludes volatile Food (44.775%) and Fuel/Utilities (Divisions 04 & 07) to measure underlying demand-pull monetary inflation on the remaining **52.200%** basket weight.
- Dynamic Missing Quote Imputation:** Temporarily unobserved items (≤ 7 days) are imputed using the geometric change of observed peer products in the same subclass: $P_{i,t} = P_{i,t-1} \times (\prod_j P_{j,t}/P_{j,t-1})^{1/n}$.

Machine Learning Inflation Nowcasting Model

- Two-Stage Hybrid Architecture:** Stage 1 aggregates daily quotes into 5 analytical streams (Food, Transport, Utilities, FX, Core). Stage 2 applies regularized Ridge regression to project month-end division index levels:

$$\hat{\beta}_k = \left(\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{y}_k$$

(3)

- with L2 regularization penalty α optimized autonomously through Leave-One-Out Cross-Validation (RidgeCV).
- 9 Momentum Features & Uncertainty Decay:** The model ingests 7-day and 30-day food velocity, 7-day transport momentum, 30-day utility inertia, 7-day exchange rate changes, month progress ratio ($\tau = t/T$), decay factor, and promotional day dummies. Uncertainty intervals contract dynamically toward month-end: $U_t = \pm 1.96 \cdot \hat{\sigma}_{\text{res}} \cdot \sqrt{(T-t)/T}$.

Monthly CPI Comparison with Official NIS Benchmark

Consumer Basket Aggregate	CSES Weight	Official NIS (MoM)	Pipeline Nowcast	Difference	Verification Status
National Headline CPI (All Items)	100.000%	+0.3470%	+0.3651%	+0.0181 pp	High Precision Alignment
Division 01: Food & Non-Alcoholic Beverages	44.775%	+0.5120%	+0.4821%	-0.0299 pp	Accurate Fresh Produce Trend
Division 04: Housing, Water, Electricity, Gas	17.084%	+0.0150%	+0.0215%	+0.0065 pp	Regulated Tariff Anchor
Division 07: Transport (Fuel & Transit)	12.180%	+0.7100%	+0.7410%	+0.0310 pp	Fuel Gazette Pass-Through
Refined Core Basket (Excl. Food & Energy)	52.200%	-0.0580%	-0.0700%	-0.0120 pp	Stable Underlying Anchor

Page 5 — Progress & Next Steps

Completed Work

- 25 automated collection routines active in production, capturing 35,792 prices daily with 99.3% reliability.
- Centralized PostgreSQL data warehouse with monthly partitioning and declarative observation constraints.
- Modular dbt Core transformations with 33 automated schema and business logic data quality checks.
- Automated daily workflow scheduled at 08:00 AM ICT via Apache Airflow and Astronomer Cosmos (runs in < 35 min).
- Zero-defect COICOP classification covering 100% of 83,180 canonical items with persistent database caching.
- Daily Jevons micro-indices and CSES 2023 weighted headline and refined core CPI calculations.
- Two-stage hybrid Ridge nowcasting model validated within ± 0.0181 percentage points of official NIS ground truth.
- Three interactive Metabase dashboards with dynamic dropdown filters for divisions and retail stores.
- Complete academic deliverable suite: 75-page graduation thesis, 20-slide defense deck, 45-page engineering handbook, and 25-page technical specification.

Current Work

- Daily surveillance of automated collection health and alert systems.
- Testing Metabase dashboard parameter filters with end users to ensure sub-second response times.
- Fine-tuning Ridge nowcasting regularization parameters as daily historical observations accumulate.
- Rehearsing graduation defense presentation with academic advisors.

Problems

Challenge Encountered	Operational Risk	Implemented Solution
Dynamic JavaScript Catalogs	Traditional downloaders receive blank pages from modern web apps.	Deployed automated browser sessions that wait for page content to finish loading before extraction.
Dual-Currency Pricing	Mixed USD and KHR prices create false price volatility.	Daily collection of the official central exchange rate and automated conversion of all prices to Cambodian Riel.
Package Size Changes (Shrinkflation)	Downsized packages appear cheaper if volume is not tracked.	Implemented 7 physical attribute checks locking weight, volume, and count into unique product identities.
AI Rate Limits and Speed	Calling external AI repeatedly causes delays and extra costs.	Built a local database cache so previously classified items are recognized instantly in under 1 millisecond.
Mathematical Index Drift	Standard arithmetic averages drift upward over time.	Applied the international standard geometric Jevons formula for all elementary price calculations.

Next Milestones

Milestone	Target Date	Responsible Party	Status	Scope of Milestone
M1: Defense Rehearsal	September 20, 2026	Academic Advisor	Scheduled	Formal rehearsal of the 20-slide graduation defense presentation.
M2: Graduation Defense	September 28, 2026	Examination Committee	Confirmed	Official thesis defense for the Bachelor of Science in Mathematics.
M3: Thesis Archive Submission	October 05, 2026	Mathematics Department / Library	Ready	Formal submission of the signed 75-page thesis and code repository.
M4: Stakeholder Handover	October 15, 2026	Technical Operations Team	In Progress	Delivery of deployment containers, operational training, and documentation.

Questions for Supervisor

1. **Agricultural Seasonality:** Does the supervisor recommend incorporating an explicit seasonal adjustment factor for volatile fresh produce, or is the current 7-day peer category imputation sufficient?
2. **Forecasting Model Window:** For ongoing reporting, should the inflation forecasting model use a fixed 90-day rolling calibration window or transition to an expanding historical window as data accumulates?
3. **Government Cloud Hosting:** What are the specific network security and server firewall requirements for deploying the containerized pipeline to the internal production infrastructure?